Catalyst Medical Education

Accelerate Learning. Improve Care.

Catalyst
MEDICAL EDUCATION

Hello. We are Catalyst.

Catalyst
MEDICAL EDUCATION

Louis Settembrino, CHCP
Founder and CEO

- 19 years' experience in Independent Medical Education (IME) across 30+ disease areas
- Expertise in strategy, instructional design, and outcomes assessment
- Long term relationships with educational partners, institutions, and faculty experts

Reshma Desai Carter, PharmD
COO

- 23 years' experience in IME across 50+ disease areas
- Expertise in outcomes measurement methodology, assessment tools, analysis, reporting, and publications
- Experience across all aspects of IME including instructional design, clinical development, accreditation standards, and outcomes

Jessica F Marshall, PhD
Vice President, Medical Strategy

- 12 years' experience in IME across 50+ disease areas
- Strong background in clinical research including literature review, gap analysis, content development
- History of managing 200+ needs assessments a year

Cathy Zehr
Director, Project Management

- 31 years' experience in IME
- Strong background in program management, meeting planning, marketing, faculty communication, and graphic design
- History of managing 50+ activities a year

Catalyst's core team members boast a collective <u>80+ years</u> of independent medical education experience!!

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Juliette Reinsel, PharmD
Manager, Educational Grants

- 5+ years' clinical experience in coordinating therapeutic care within 1000+ bed academic teaching hospital, corresponding with providers, nurses, and social workers
- Expertise in conducting root cause analysis to determine underlying gaps in care and knowledge ex. "fix-it pharmacist"
- Clinical perspective of therapeutic landscapes and ability to identify educational gaps

Nicole Grassano, CHCP
Manager, Educational Grants

- 10 years' of experience in IME business development and continuing medical education project execution.
- Experience across 20+ therapeutic areas identifying clinical needs and aligning with educational design.
- Established relationships and providing white glove client service.

Alexandra Marsillo, PhD
Director, Medical and Scientific Services

- 6 years' of diverse experience in IME, with a specialized focus on oncology
- Demonstrated success in directing grant and education material development to achieve impactful outcomes, aligning stakeholder objectives with learners' educational needs
- Background in clinical research and higher education, emphasizing scientific understanding and effective communication

Julie Blum, PhD
Director, Medical and Scientific Services

- 13 years' experience in IME, adept at designing, creating, implementing, and managing innovative medical education
- Expertise in crafting content that resonates with clinical and patient audiences and aligns with stakeholder objectives
- History of managing 30+ programs per year in dozens of disease states

Catalyst's team members boast a collective 100+ years of independent medical education experience!!

Why Catalyst?

- Amortize investments in content development (most expensive budget item)
- Develop an “effective” core curriculum (strategic framework, including ALPs)
- Utilize multiple audience generation partners
- **Provide valuable outcomes data that informs <u>internal stakeholders</u> and defines opportunities for future education**

ALP = Adult Learning Principles

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We collaborate with trusted institutions and educational partners including:

- Accredited Providers
- Medical Societies
- Audience Generation Partners
- Patient Education Platforms
- Global Education Partners
- Technology Partners

Planning Methodology

Phase	P II/III	NDA	Launch	Post Launch
Goal	Integrated Planning	Knowledge Gaps	Competence Gaps	Performance Gaps
Strategy	Establishing Needs	Expert Driven Methods	Learner Driven Methods	System Driven Methods
Methods	• Gap Analysis • Needs Assessment • Pilots • Change Readiness	• Lecture Based • Symposia • Traditional Web • Print	• Case Based • Self Directed Education • PBLI – Stage A & B	• Quality Improvement • Barrier Mitigation • PBLI – Stage C
Targets	S	S	S + PC	S + PC + AH

PBLI = Practice Based Learning and Improvement; S = Specialists; PC = Primary Care; AH = Allied Health

Activity Formats

Live*

Roundtables
Symposia
Regional Meetings
Regional Workshops
Summit Meetings
Hands-on Training
Wet Labs
Webcasts

Web

Single Activities
Web Series
Microsites
Learning Centers

Print

Monographs
Supplements

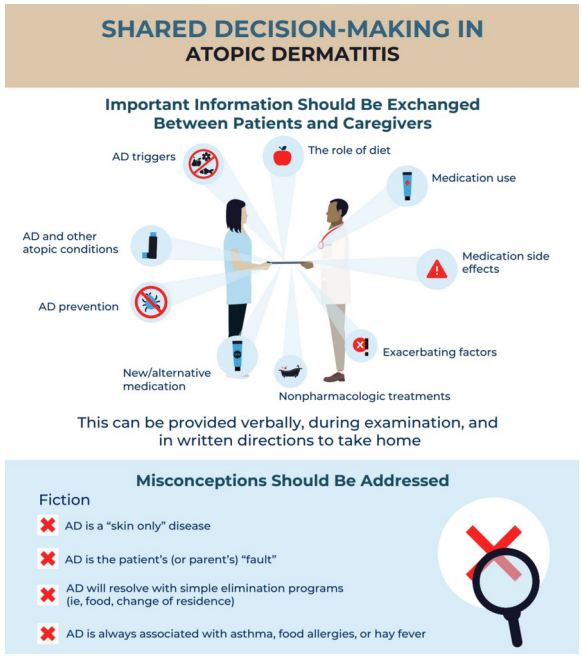
Other

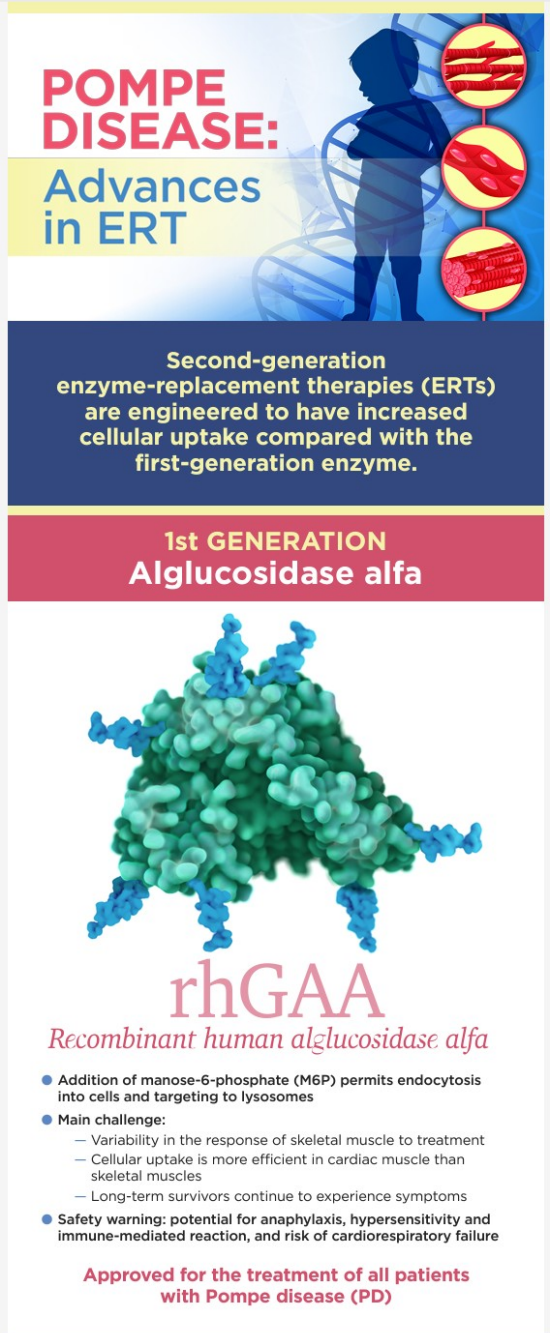
MOA Animations
Mobile Apps
Residents/Fellows
Patient Education
MSL Education
Needs Assessments

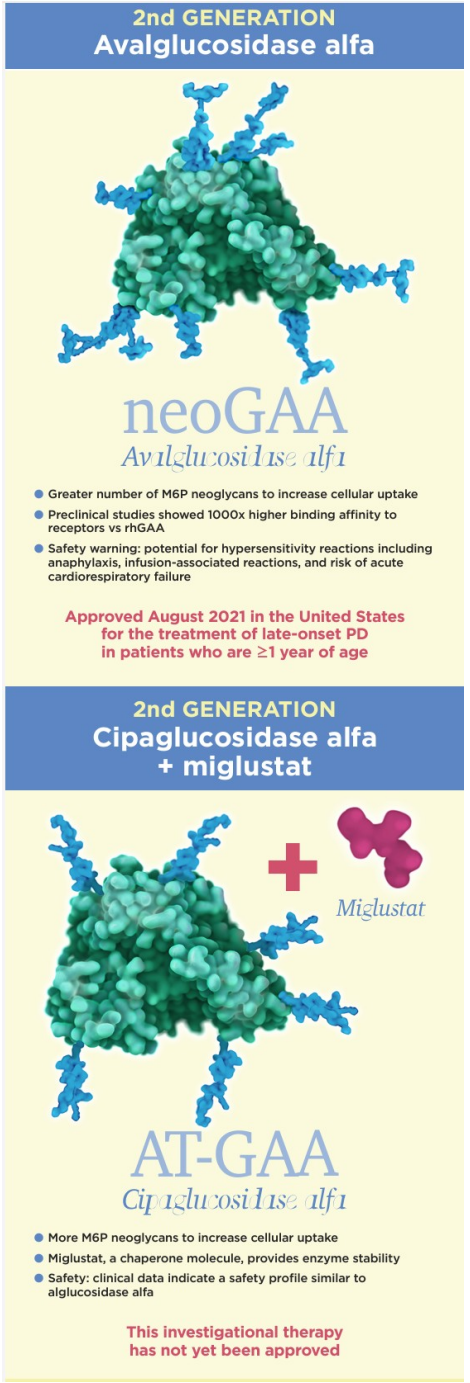
*Most live options can be developed as virtual or hybrid meetings

Reinforcing Components

Example Infographics and Translations







Global Opportunities

- Implement global needs analysis to develop geographically focused “blueprint” for education
- Develop a global core curriculum that
  - Addresses gaps identified by region
  - Lifecycle/approval status
  - Regulatory/reimbursement environment
  - Economic conditions
  - Cultural issues
- Global symposia
- Global regional/local meetings
- Global enduring education

Strategy and coordination lead to improved effectiveness of education and cost savings!

Educational Examples

NEW REEL

Learning Centers

Pompe Disease: https://pompecme.com/

MASH: https://mash-ce.com/

The IMPACT Initiative: https://impact-smd.com/

Distributed nationally or globally, via a journal package, with the option to offer content and resources in various languages, as appropriate (see resources page for examples).

Pharmacy Education

https://www.powerpak.com/course/preamble/120595

We have a successful partnership with the University of Tennessee College of Pharmacy to provide both live (primarily webcasts) and enduring media.

Animations

https://catalystmeded.com/animation

MOA animations have been developed to help learners visualize key concepts for both live and on-demand initiatives. This specific animation was used in a series of symposia (ICIEM, WORLD Symposium, ACMG, etc), and correlating enduring archives. For a hemophilia example, see https://catalystmeded.com/animation2

Patient Learning Portal

https://catalystmeded.com/PT-P

This comprehensive patient education portal utilizes infographics and translations to ensure content is literacy level appropriate and easily digestible for a patient audience. The site and materials are designed to be mobile friendly. Distribution is multifaceted and targets both relevant HCPs and patients directly.

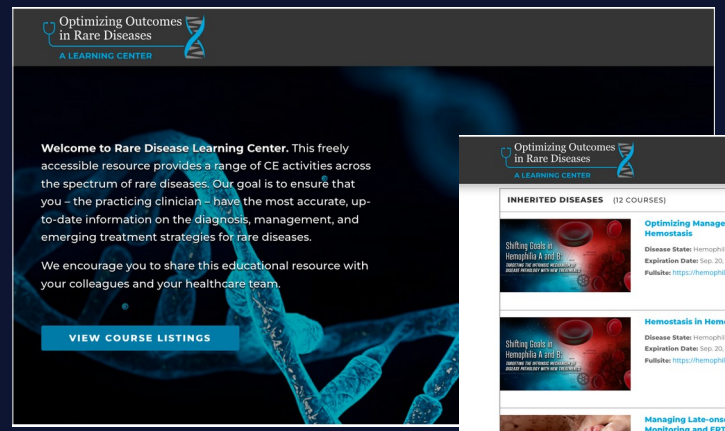
Patient sites pair well with HCP-focused learning centers, to ensure that correlating educational messages and information reach both audiences.

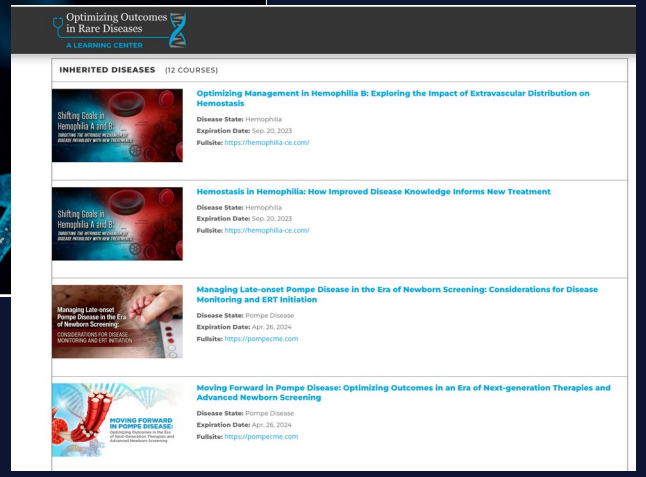
Rare Education Expertise

Rare Ed Learning Center (https://rare-ed.com/)

Portal allowing access to all our rare disease CE activities, allowing learners to explore the spectrum of rare diseases

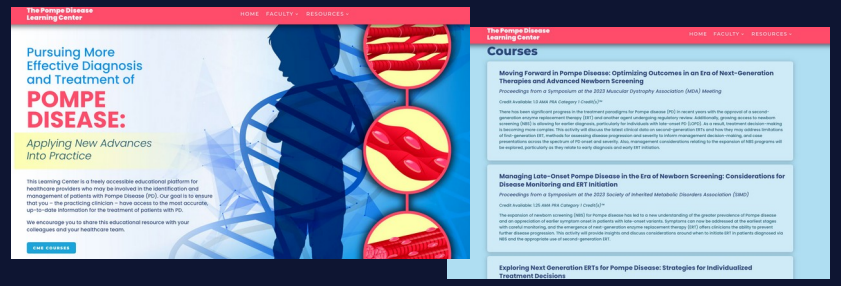
- 30+ courses posted as of February 2025
- Courses include Pompe disease, Fabry disease, acid sphingomyelinase deficiency, GM2 gangliosidosis, spinal muscular atrophy, chronic inflammatory demyelinating polyradiculoneuropathy
- Each course hosted and marketed individually as part of our partnership with NexusMedX





Lysosomal Storage Disorder Experience

- Pompe disease
  - 12+ CE activities (and counting): live/livestream/rebroadcasts; enduring lecture-based; enduring case-based; clinical animations; learning centers
  - Learning Center: https://pompecme.com/



- Fabry disease
- Acid sphingomyelinase deficiency (ASMD)
- Mucopolysaccharidosis type 1 (MPS 1)

Other Rare Inherited Disease Experience

- Hematology: hemophilia; sickle cell disease
- Dermatology: hereditary angioedema
- Metabolic/endocrine disorders: fibrodysplasia ossificans progressive
- Nephrology: monogenic kidney stone disorders
- Hepatology: acute hepatic porphyria

Audience Generation

Overview

For these activities, Catalyst will collaborate to design and implement a comprehensive distribution plan that will leverage various partner networks.

They will drive recruitment for CME activities through digital marketing, print, and social media channels.

Methods

- Faculty networks
- Banner campaigns on key websites/journals
- Banner campaigns in eTOC or eNewsletter alerts sent to subscribers
- Multiple e-blasts to target audience via specialty (i.e., “opted in”) databases
- Activity posting throughout media channels and websites (e.g. recommended courses, program carousel, or badge ads space to drive traffic with enhanced placement options)
- LinkedIn advertising and Social Media Channels
- Dedicated paid Google ad word campaign

Outcomes Snapshot

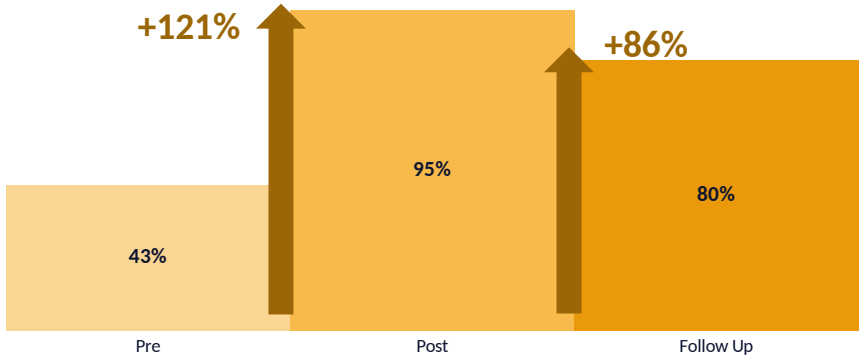
Satisfaction

96% Found VALUE

Learners indicate they gained new strategies, skills, and/or information relevant to their practice.

- 4,433 total learners (88 live and 4,345 enduring).
- Most learners were Physicians at 66%, followed by Nurse/nurse practitioners at 13%.
- Satisfaction was high across the board: Faculty scored 3.9/4.0, the Educational Objectives scored 3.8/4.0, and the Activity Components scored 3.8/4.0.

Increase in Knowledge/Competence from Baseline



Pre n=613 CL at 95% CI = +/-3% Post n=605 CL at 95% CI = +/-3% Follow Up n=81 CL at 95% CI = +/-9%

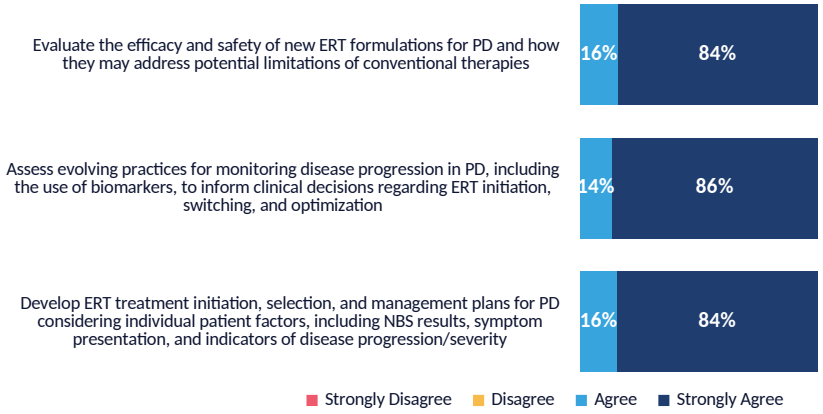
Practice Implications

90% Intend to take action!

Learners indicate that they plan to implement changes into practice due to the training.

- 69% said they would change their practice.
- 52% of learners said that they would seek additional information.
- 2,014 potential yearly patient visits impacted.

As a result of my participation in this activity, I am able to:



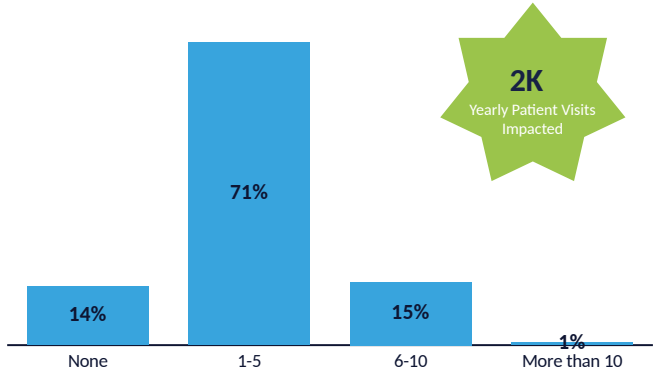
Post n=604 CL at 95% CI = +/-3%

Educational Gains

+121% Increase in KNOWLEDGE/ COMPETENCE

- Knowledge: Immediate knowledge saw a +162% lift; longer-term saw a +120% lift.
- Competence: Immediate competence saw a +91% lift; longer-term saw a +62% lift.
- Combined Knowledge/Competence: Immediate saw a +121% lift; longer-term saw a +86% lift.

Approximately how many patients with Pompe disease (PD) do you see each year?



Pre n=569 CL at 95% CI = +/-3%

Educational Synopsis: Summary

Gaps/Learning Objectives	Curricula/Instructional Methods	Learning Features
Gap: Due to its rarity, lack of NBS, and overlap with other inherited conditions, diagnosis of MLD is often missed or delayed, precluding early intervention, particularly considering gene therapy • **Describe** the genetic etiology, pathophysiology, clinical manifestations, and natural history of MLD as it relates to the necessity of early intervention • **Apply** appropriate genetic testing, family history assessments, and symptom assessments to facilitate early identification and intervention in MLD	**Introduction** • Genetic etiology, enzymatic deficiency, clinical effects • Genotype/phenotype spectrum and subtypes, including morbidity and mortality • Urgency for early identification considering the availability of gene therapy **Facilitating Early Diagnosis in the Emerging Disease-modifying Treatment Era** • Stages and related presentations considering late-infantile, juvenile, adult subtypes • Clinical comparison/overlap with other LSDs and leukodystrophies • Impact of delayed and incorrect diagnosis of MLD – focus on missed opportunities to initiate emerging gene and other therapies in earlier phases of disease progression **Clinical Case Challenge: Achieving an Early and Accurate MLD Diagnosis** • Implementing screening, clinical assessments, genetic testing, and interpretation • Addressing diagnostic caveats (e.g., enzyme pseudodeficiency) **Designed to close gaps in: *Knowledge*, *Competence*, *Performance (self-reported)***	**Interactive features:** **e-polling knowledge/case-based competence** **Reinforcing features:** • **Infographic 1.** Assessment and Diagnostic Algorithm for MLD • Suggested readings

Gap: With MLD gene therapy and other emerging strategies on the horizon, HCPs require foundational education on the most up-to-date clinical and real-world data on efficacy and safety
• **Assess** clinical trial and real-world data on the efficacy and safety of MLD gene therapy, and other emerging treatments
	Evaluating Gene Therapy and Other Emerging MLD-specific Therapies
	• Overview: supportive care treatment options and their limitations
	• Atidarsagene autotemcel
	  o ex-US and US regulatory approval and clinical indications
	  o Overview of ex-vivo gene therapy technology, preparation, administration
	  o Clinical trial data: efficacy and safety
	  o Real-world and longer-term data
	• Emerging ERT; role of HSCT; other investigational disease-specific strategies
	Designed to close gaps in: *Knowledge*
		Interactive features:
		e-polling knowledge
		Reinforcing features:
		• **Infographic 2.** MLD Gene Therapy
		• Suggested readings by topic

Gap: HCPs lack awareness of practical considerations for integrating into emerging gene therapy into comprehensive care paradigms, including patient eligibility • **Develop** strategies that support the early and appropriate integration of gene therapy into MLD clinical scenarios • **Review** the implications of MLD gene therapy on the concept of NBS	**Clinical Case Challenge: Optimizing Treatment Outcomes Considering Gene Therapy** • Patient eligibility considering varying MLD subtypes and stages within those subtypes • Clinical scores and measurement tools used assess stage • Gene therapy preparation and administration considerations • Assessing effectiveness, adverse events, tolerability • Potential role of other emerging strategies via clinical trial enrollment • Implications on NBS, cascade testing, etc. **Designed to close gaps in: *Competence*, *Performance (Self-reported)***	**Interactive features:** **e-polling case-based competence** **Reinforcing features:** • Suggested readings by topic

The Value of Working with Catalyst

- Unparalleled Service
- Strategic Instructional Design
- Cost-effective & Outcomes-driven Education
- Extensive Experience & Track Record of Success

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Thank You

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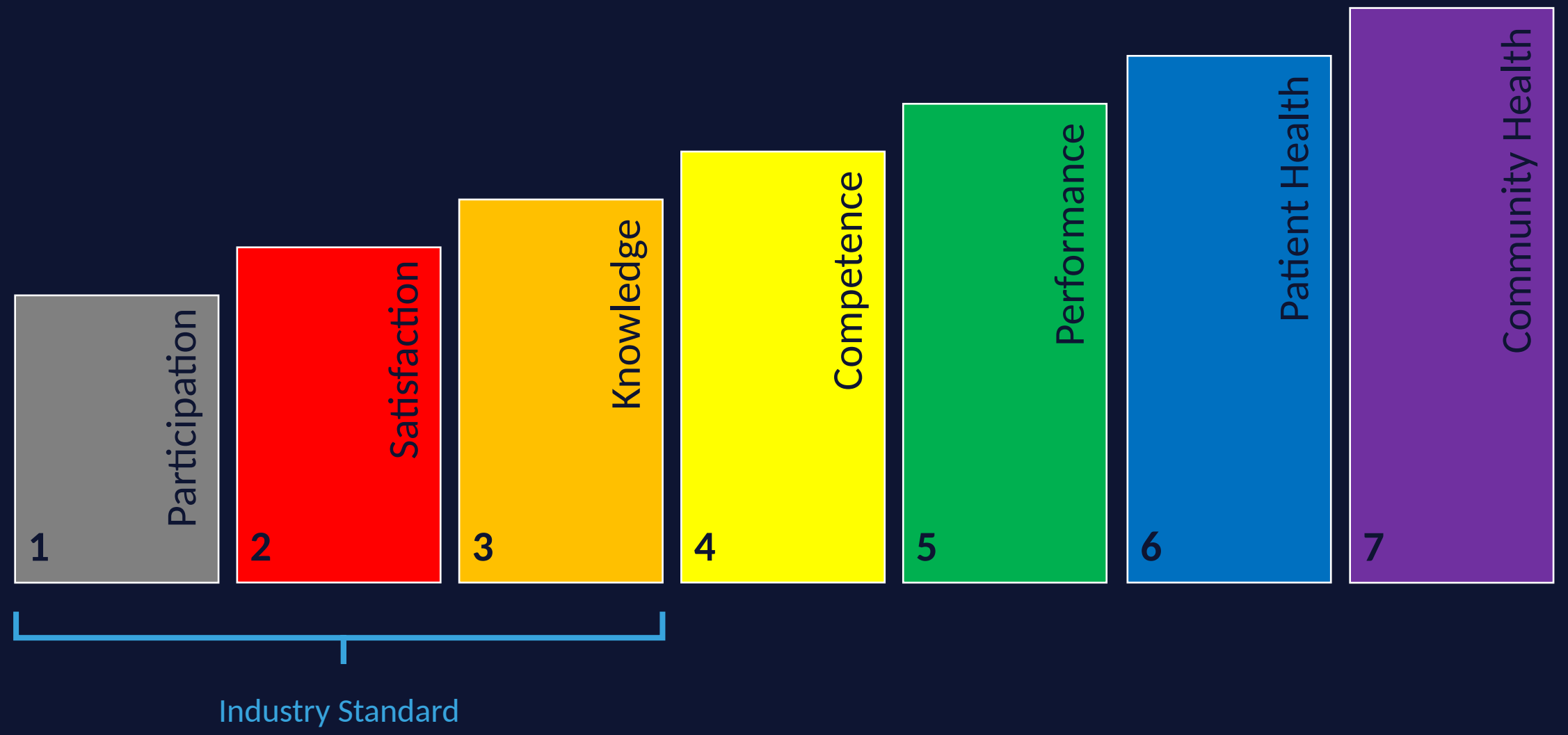
Planning Methodology: Core Focus

Phase	P II/III	NDA	Launch	Post Launch
Goal	Integrated Planning	Knowledge Gaps	Competence Gaps	Performance Gaps
Strategy	Establishing Needs	Expert Driven Methods	Learner Driven Methods	System Driven Methods
Methods	• Gap Analysis • Needs Assessment • Pilots • Change Readiness	• Lecture Based • Symposia • Traditional Web • Print	• Case Based • Self Directed Education • PBLI – Stage A & B	• Quality Improvement • Barrier Mitigation • PBLI – Stage C
Targets	S	S	S + PC	S + PC + AH

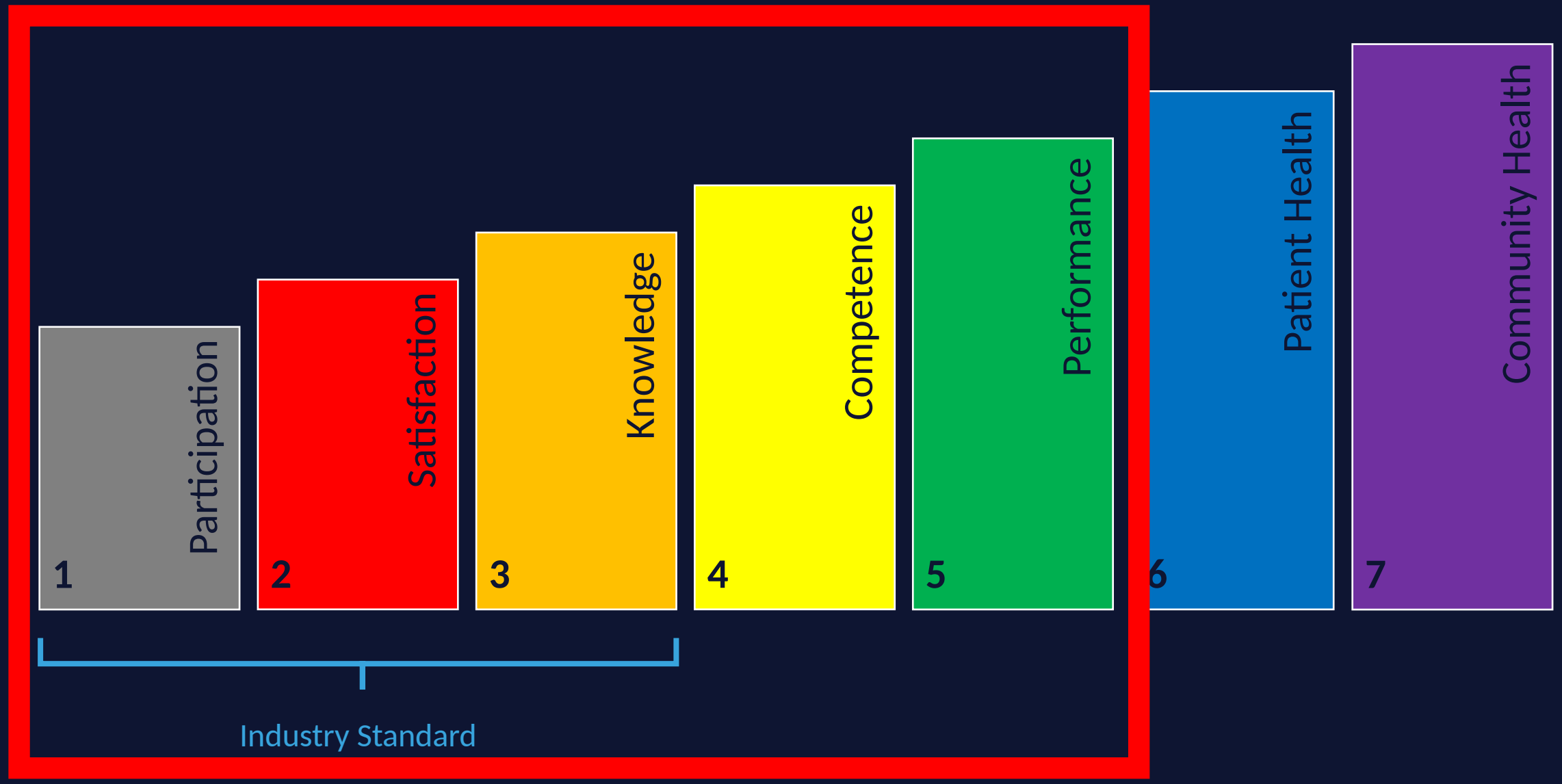
PBLI = Practice Based Learning and Improvement; S = Specialists; PC = Primary Care; AH = Allied Health

Outcomes Measurement Levels



Moore DE. Achieving desired results and improved outcomes: integrating planning and assessment throughout learning activities.

Outcomes Measurement: Core Focus



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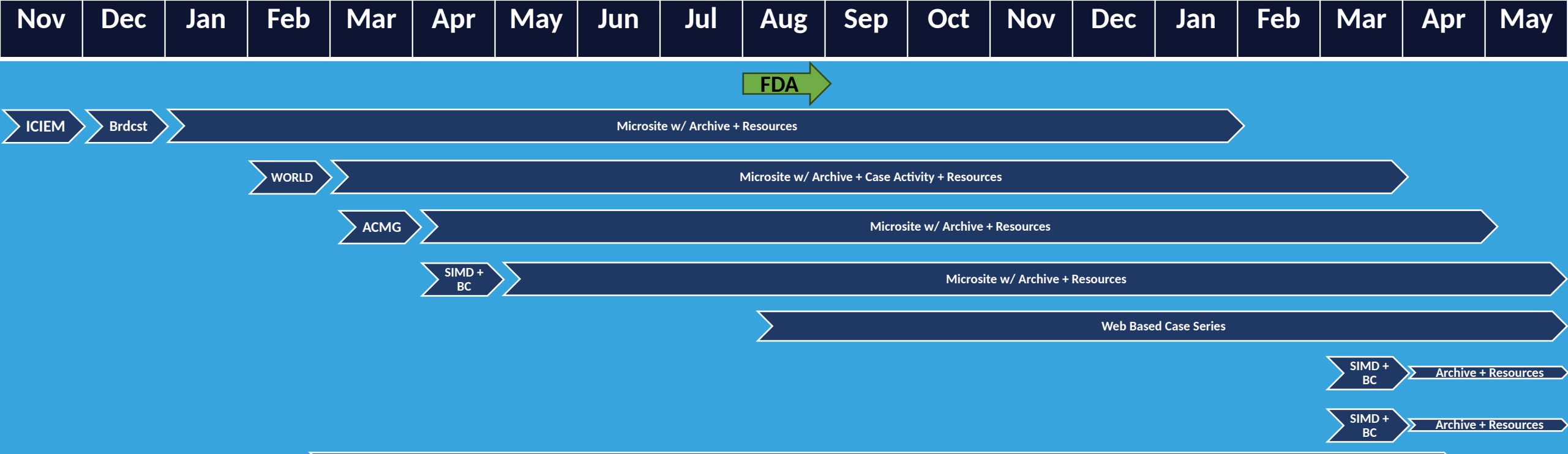

Learning Center at: https://pompecme.com/

The Learning Center hosts all enduring media listed above and reinforcing resources listed below.

Resource Reinforcing List

- Infographic - Enhancing Pompe Disease Recognition to Prompt Diagnosis
- Infographic - Pompe Disease: Advances in ERT
- MOA Animation – downloadable video
- FAQ Resource (Updated → ICIEM, WORLD, ACMG, SIMD)
- Access to available guidelines/ recommendations (Updated → ICIEM, WORLD, ACMG, SIMD, etc)
- Suggested Readings (Updated → ICIEM, WORLD, ACMG, SIMD)

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Pompe Disease Initiative Gantt Chart

CME

Resources

Educational Synopsis: Gap 1 (Diagnosis)

Gaps/Learning Objectives	Curricula/Instructional Methods	Learning Features
Gap: Due to its rarity, lack of NBS, and overlap with other inherited conditions, diagnosis of MLD is often missed or delayed, precluding early intervention, particularly considering gene therapy • **Describe** the genetic etiology, pathophysiology, clinical manifestations, and natural history of MLD as it relates to the necessity of early intervention • **Apply** appropriate genetic testing, family history assessments, and symptom assessments to facilitate early identification and intervention in MLD	**Introduction** • Genetic etiology, enzymatic deficiency, clinical effects • Genotype/phenotype spectrum and subtypes, including morbidity and mortality • Urgency for early identification considering the availability of gene therapy **Facilitating Early Diagnosis in the Emerging Disease-modifying Treatment Era** • Stages and related presentations considering late-infantile, juvenile, adult subtypes • Clinical comparison/overlap with other LSDs and leukodystrophies • Impact of delayed and incorrect diagnosis of MLD – focus on missed opportunities to initiate emerging gene and other therapies in earlier phases of disease progression **Clinical Case Challenge: Achieving an Early and Accurate MLD Diagnosis** • Implementing screening, clinical assessments, genetic testing, and interpretation • Addressing diagnostic caveats (e.g., enzyme pseudodeficiency) **Designed to close gaps in: *Knowledge*, *Competence*, *Performance (self-reported)***	**Interactive features:** **e-polling knowledge/case-based competence** **Reinforcing features:** • **Infographic 1.** Assessment and Diagnostic Algorithm for MLD • Suggested readings

Educational Synopsis: Gap 2 (Treatment)

Gaps/Learning Objectives	Curricula/Instructional Methods	Learning Features

Gap: With MLD gene therapy and other emerging strategies on the horizon, HCPs require foundational education on the most up-to-date clinical and real-world data on efficacy and safety
 	
• **Assess** clinical trial and real-world data on the efficacy and safety of MLD gene therapy, and other emerging treatments
	Evaluating Gene Therapy and Other Emerging MLD-specific Therapies
	• Overview: supportive care treatment options and their limitations
	• Atidarsagene autotemcel
	  o ex-US and US regulatory approval and clinical indications
	  o Overview of ex-vivo gene therapy technology, preparation, administration
	  o Clinical trial data: efficacy and safety
	  o Real-world and longer-term data
	• Emerging ERT; role of HSCT; other investigational disease-specific strategies
	
	Designed to close gaps in: *Knowledge*
		Interactive features:
		e-polling knowledge
		
		Reinforcing features:
		• **Infographic 2.** MLD Gene Therapy
		• Suggested readings by topic

Educational Synopsis: Gap 3 (Clinical Applications and Implications)

Gaps/Learning Objectives	Curricula/Instructional Methods	Learning Features
Gap 3: HCPs lack awareness of practical considerations for integrating into emerging gene therapy into comprehensive care paradigms, including patient eligibility • **Develop** strategies that support the early and appropriate integration of gene therapy into MLD clinical scenarios • **Review** the implications of MLD gene therapy on the concept of NBS	**Clinical Case Challenge: Optimizing Treatment Outcomes Considering Gene Therapy** • Patient eligibility considering varying MLD subtypes and stages within those subtypes • Clinical scores and measurement tools used assess stage • Gene therapy preparation and administration considerations • Assessing effectiveness, adverse events, tolerability • Potential role of other emerging strategies via clinical trial enrollment • Implications on NBS, cascade testing, etc. **Designed to close gaps in: *Competence, Performance (Self-reported)***	**Interactive features:** **e-polling case-based competence** **Reinforcing features:** • Suggested readings by topic